CSC / AI Course — Expert System Mini Project (Rule-Based Inference)

■ Project Goal

You will build a working expert system using forward chaining, where a computer draws conclusions from a set of facts and IF–THEN rules. You are provided with a partially completed project folder. Your job is to complete the missing parts, add your own rules, and make the system reason correctly.

1. Setup

1. Download and unzip the file: expert_system_teaching_version.zip
2. Inside, you'll see the project structure with incomplete files (engine.py, main.py, and one example rule).
3. Open the folder in your preferred code editor (VS Code, PyCharm, or IDLE).

2. Run the Program

1. Ensure Python 3.8+ is installed. Check with: python3 --version
2. Navigate to the project folder in your terminal or command prompt.
3. Run the program: python3 main.py
4. You should see prompts like 'Is portability important? (y/n):' — after implementing, the system will infer recommendations.

3. What You Need to Do

You will complete three main files:

a) engine.py — Implement reasoning logic:

- Implement can_fire(rule): return True if all antecedents are satisfied.
- Implement run(): main forward chaining loop — repeatedly apply rules until no more fire.
- Implement conclusions(): collect all derived recommendations/specs.

b) kb/laptop_rules.json — Expand the knowledge base:
Currently only one rule is given. Add at least nine more IF–THEN rules.

c) main.py — Complete the program:

- Load rules using load_rules().
- Collect user facts via input questions.
- Run the inference engine and print results (recommendations, fired rules).

4. Example Output (After Completion)

```
Is portability important? (y/n): y
Do you need long battery life? (y/n): y
Is your budget high? (y/n): y
```

```
> Recommendation: premium_ultrabook
> Explanation: derived from rule 'Premium Ultrabook'
```

5. Testing

1. Add test cases under the /tests folder.
2. Each test should include input facts and expected output.
3. Example: Input: ['budget_high', 'portable', 'long_battery'] → Expected: recommend:premium_ultrabook.

6. Grading (100 points)

Component	Points
Inference engine implemented correctly (engine.py)	40
Knowledge base: ≥10 complete rules	20
Correct reasoning and output	20
Code clarity & structure	10
Report / explanation	10
Bonus: explanations, uncertainty handling	+10

7. Hints and Tips

- Start small: get one rule to fire correctly, then add more.
- Use print statements to debug and trace your reasoning steps.
- Draw a simple flowchart of how facts lead to conclusions.
- Avoid hardcoding logic — your inference should work for any rule set.